HOME PRO SERIES · 12 V LIFEPO₄ BATTERY

12.8 V
100 Ah

Deep-cycle lithium energy storage · Model TLF-1208100

A 1.28 kWh $LiFePO_4$ battery pack built on 3.2 V 100 Ah prismatic cells (4S1P) and a smart BMS with cell balancing and multi-protocol communication. Made for home inverters, rooftop solar and power backup in homes, shops, offices and godowns.



LiFePO₄ Prismatic | 4S 1P | Smart BMS | CAN · RS485 · UART · BT | MS Casing

12.8 V	100 Ah	1.28 kWh	≥2500
NOMINAL VOLTAGE	RATED CAPACITY	NOMINAL ENERGY	CYCLES @ 90% DOD

BUILT FOR DEEP, DAILY CYCLING

Prismatic LiFePO₄ cells
3.2 V 100 Ah cells in a 4S1P layout — a stable, cobalt-free chemistry.

Smart BMS
Over-charge, deep-discharge and short-circuit protection with active cell balancing.

Multi-protocol comms
CAN, RS485, UART and Bluetooth for inverter integration and monitoring.

High discharge capability
100 A continuous (1.28 kW) and 150 A pulse (2 s) for demanding loads.

Fast recharge
Supports 50 A rapid charging (0.5C) and 1.15 kWh usable energy at 90% DoD.

Rugged steel casing
Powder-coated MS enclosure, 260 × 220 × 210 mm, 13 kg.

MADE FOR HOMES & SMALL BUSINESSES

Homes & villas | Shops & showrooms | Small offices | Godowns & warehouses | Clinics & salons

Technical Specifications

12.8 V 100 Ah $LiFePO_4$ Battery Pack · TLF-1208100

ELECTRICAL PERFORMANCE

Module configuration	4S 1P	
Nominal voltage	12.8 VDC	
Operating voltage range	11 – 14.6 VDC	
Installed energy	1.28 kWh	
Usable energy	1.15 kWh	@ 90% DoD

Rated charge current	20 A	0.2C
Rapid charge current	50 A	0.5C
Rated discharge current	100 A	
Pulse discharge current	150 A	@ 2 s
Rated discharge power	1.28 kW	

MECHANICAL

Cell	Prismatic, 3.2 V – 100 Ah
Housing material	MS (mild steel) casing
Dimensions (L × B × H)	260 × 220 × 210 mm
Dimension tolerance	±5 mm
Weight	13 kg ±5%
Terminal	C-terminal connector
IP grade	NA

BMS & COMMUNICATION

Protection	Deep discharge & over-charge
Short circuit	Protected
Cell management	Cell balancing
Temperature	Temperature compensation
Communication	CAN / RS485 / UART / Bluetooth

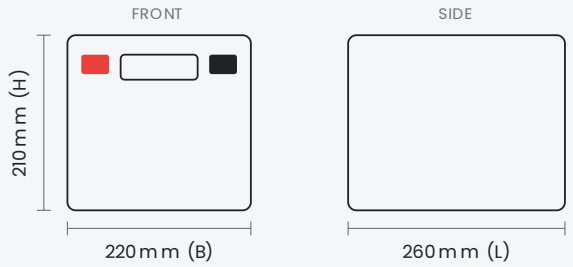
OPERATING TEMPERATURE

Charging	0 – 45 °C
Discharging	0 – 55 °C

CYCLE LIFE

Cycle life	≥ 2,500 cycles
Test conditions	90% DoD · 25 ±3 °C · 0.2C / 1C

DIMENSIONS



STORAGE CONDITIONS

DURATION	TEMPERATURE
Up to 1 year	-20 to 25 °C
Up to 6 months	25 to 35 °C
Up to 3 months	35 to 45 °C
Less than 1 week	45 to 60 °C

Store at 50 ±5% SOC, humidity < 60 ±25% RH, non-condensing.

PRODUCT & SAFETY NOTE

- Use only with a compatible $LiFePO_4$ charger / inverter and within the specified operating limits.
- Final product certification and transport compliance depend on the exact production configuration.
- Specifications and warranty conditions may vary with load profile and operating conditions.
- Touch Energy reserves the right to change specifications without prior notice.